🛰️ 1. NDVI / Satellite Imagery Datasets

Dataset	Description	Direct Link
Sentinel-2 Sample NDVI	Pre-processed Sentinel-2 image (4 bands including NIR & RED) — perfect for testing your NDVI code.	📂 Sentinel Sample .tif
NASA MODIS NDVI Collection	Global NDVI data (monthly averages, 1km resolution).	🌍 MODIS NDVI
USGS EarthExplorer (Free account)	Download satellite data (Sentinel-2, Landsat-8, etc.) for any region.	🌎 EarthExplorer
Copernicus Open Access Hub	Raw Sentinel data (use NIR and RED bands for NDVI).	🛰️ Copernicus Hub
Google Earth Engine Public Data	Access NDVI datasets instantly via code (Python API available).	🪐 Earth Engine Datasets

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🌲 2. LiDAR / 3D Forest & Terrain Datasets

Dataset	Description	Direct Link
OpenTopography – Sonoma County, CA	Free high-resolution LiDAR dataset (used in our MVP).	📂 Sonoma Dataset (.las)
NEON Forest Structure LiDAR Data (US)	Full forest canopy + terrain LiDAR point clouds — ideal for research.	🌿 NEON LiDAR Data Portal
USGS 3D Elevation Program (3DEP)	LiDAR-based terrain data across the US (DEM + point cloud).	🗺️ USGS 3DEP
ISPRS Benchmark LiDAR Dataset	Academic dataset for point cloud segmentation and classification.	📊 ISPRS Vaihingen Dataset
OpenTopography India (if regionally needed)	Local LiDAR projects (limited but expanding).	🇮🇳 OpenTopography India

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🌿 3. Vegetation Health / AI Training Datasets

Dataset	Description	Direct Link

Kaggle – Forest Health Monitoring	NDVI and multispectral images labeled as *healthy* or *stressed*.	🌳 Forest Health Dataset
Kaggle – Crop and Weed Detection	Drone imagery dataset for plant classification.	🌾 CropWeed Dataset
Global Forest Watch Open Data	Deforestation & vegetation change data from 2001–2023.	🌍 GFW Open Data Portal
PlantVillage Dataset (Leaf Health)	50,000+ labeled plant leaf images — great for CNN training.	🍃 PlantVillage Dataset
UAV-based Forest Dataset (MDPI 2024)	Aerial drone imagery with LiDAR + RGB fusion (paper dataset).	📸 UAV Forest Dataset – MDPI

⚙️ 4. Recommended Small Datasets for Quick Testing

If you're just validating your MVP pipeline:

Type	Direct Command	Note
LiDAR	`wget https://opentopography.s3.sdsc.edu/lidar/demonstration/Sonoma_CA.las -O data/forest_sample.las`	Already used in your MVP
NDVI	`wget https://github.com/chrieke/awesome-sentinel/raw/main/sample_data/sentinel_image.tif -O data/satellite_image.tif`	Small, ready-to-use .tif image

🔬 How to Use These in Your Project

1. **Download one LiDAR file** (`.las`) → save in `data/forest_sample.las`
2. **Download one NDVI / satellite image** (`.tif`) → save in `data/satellite_image.tif`

Run:

```
python src/main.py
```

3.
4. Your pipeline will:
    - Load LiDAR → Compute canopy height + density
    - Compute NDVI → Generate vegetation health index
    - Simulate AI prediction → “Healthy / Stressed”